RX

RetailX

Retail Store Sales Analytics

Data Visualization & Analytics — Capstone 2

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Team ID

SectionB-G13-RetailX

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Sector: Retail

Context & Problem Statement

Sector Context

Retail is one of the most margin-sensitive industries. Success depends on inventory optimization, targeted marketing, and rapid response to demand shifts.

Decision-maker:
Retail Operations Manager — responsible for inventory, sales strategy, and customer engagement.

Problem Statement

How can we optimize revenue and customer retention by identifying high-performing product categories, seasonal sales trends, and the impact of promotional discounts on customer purchasing behavior?

Map Trends

Identify category & seasonal sales drivers

Validate Discounts

Measure statistical impact of promotions on spend

Enable Decisions

Deliver a decision-ready Tableau dashboard

Scope: 12,575 transactions · Jan 2022–Aug 2024 · 8 categories · Omnichannel

Data Engineering · Source → Pipeline

Dataset Overview

Source	Simulated Retail Operations Dataset
Rows	12,575 transaction records
Columns	11 (source) → 14 (processed)
Period	January 2022 – August 2024
Channels	Online & In-store (Omnichannel)
Categories	8 product categories

Python ETL Pipeline

1. **Standardize:**
   Column headers normalized to snake_case
2. **Null Checks:**
   No nulls in total_spent or category
3. **Feature Engineering:**
   Extracted month, day_of_week, year from transaction_date

Data Dictionary — Key Columns

`transaction_id`	Primary key for analysis
`customer_id`	RFM & retention modeling
`total_spent`	Primary revenue KPI
`category`	Performance segmentation
`discount_applied`	Flag for sensitivity analysis

KPI & Metrics Framework

Total Revenue

₹15,66,287

Formula: SUM(total_spent)

Why: Overall business health

Avg Order Value

₹124.56

Formula: Revenue ÷ Transactions

Why: Per-visit spending power

Repeat Purchase Rate

100%

Formula: (Repeat Customers ÷ Total) × 100

Why: Customer loyalty metric

Seasonal Index

Nov–Dec

Formula: Monthly Rev ÷ Avg Monthly Rev

Why: Identifies demand peaks

Discount Effectiveness

33.55%

Formula: Count(Discounted) ÷ Total Txns

Why: Promotion impact measure

All KPIs computed in Python notebook 04_kpis.ipynb and visualized via Tableau Public

Key Insights from Exploratory Analysis

01 Customer Concentration Risk

Only 25 unique customers drive 12,575 transactions. Losing even 2–3 clients could drop revenue by 15%+.

02 Balanced Category Portfolio

All 8 categories contribute 11–13% each — Butchers (13.4%) leads, Milk (11.6%) trails. No single-category dependency.

03 Weekend Sales Spike — Fri & Sun Lead

Friday and Sunday see the highest transaction volumes. Weekday performance is ~15% lower on average.

04 Equal Online vs In-store Split

Online and In-store channels each contribute ~50% of total sales, requiring synchronized inventory management.

05 January Spend Peaks, Then Stabilizes

Customer spending hits ₹7K in January, drops to ₹5K–5.5K from Feb onwards — post-holiday demand normalization.

06 Uniform Payment Behavior

Cash (34.64%), Digital Wallet (32.69%), Credit Card (32.67%) — almost perfect thirds, enabling flexible payment promotions.

Advanced Statistical Analysis

ANOVA — Category Revenue Variance

Hypothesis

H_0: All 8 categories generate equal revenue
H_1: At least one category significantly differs

Result

p-value < 0.001 — Highly Significant

Business Interpretation

Category performance differs significantly. Inventory investment should be weighted proportionally — overstock in Milk; prioritize Butchers & Electric.

Library: scipy.stats.f_oneway

T-test — Discount Impact on Spend

Hypothesis

H_0: Discounts do not affect per-transaction spend
H_1: Discounted transactions have higher spend

Result

p-value ≈ 0.47 — Not Significant

Business Interpretation

Discounts drive transaction volume but NOT basket size. Current promotions erode margins without increasing AOV. Pivot from price-slashing to bundling.

Library: scipy.stats.ttest_ind

Tableau Dashboard Walkthrough

Retail Dashboard

- Total Sale: ₹15,66,287
- Total Transactions: 12,575
- Avg Order Value: ₹124.56
- Discount Rate: 33.55%
- Yearly Sales Trend (2022–2025)
- Day-wise & Monthly Sale charts
- Location: Online vs In-store

Customer Analysis

- 25 Total Customers
- Avg Spend/Customer: ₹125
- Discount Usage Rate: 33.55%
- Payment Method Breakdown
- Top 5 Customers ranked
- Customer Spending Trend over Time
- Key retention risk insights

Product & Category Dashboard

- 8 Total Categories
- Avg Price Per Unit: ₹23.37
- Total Units Sold: 66,880
- Total Revenue: ₹1.57M
- Category-wise Sales (bar + donut)
- Revenue by Payment Method
- Filters: Category Set, Year

Interactive Filters Available: Product Category · Year · Store Location | Data available through Jan 2025

Strategic Recommendations

1 VIP Loyalty Club

Insight: 25 customers drive all revenue

Launch 'RetailX Platinum' — exclusive perks, early access, and dedicated account managers for top 25.

Impact: Reduce churn risk by 40%+

2 Promotion Redesign

Insight: Discounts don't raise basket size (p=0.47)

Replace flat discounts with 'Bundle & Save' combos. E.g., Butchers + Electric paired bundles.

Impact: Increase AOV by 10–15%

3 Staffing Optimization

Insight: Fri & Sun are peak sales days

Shift 30% of weekday labor hours to Fri–Sun. Use predictive scheduling for seasonal peaks (Nov–Dec).

Impact: Reduce labor waste by 20%

4 Financing Partnerships

Insight: Electric buyers prefer credit cards

Partner with banks for zero-cost EMI on Electric & Computer purchases above ₹500.

Impact: Unlock premium-ticket segment

5 Real-Time Inventory Sync

Insight: 50/50 Online vs In-store parity

Implement live stock-visibility API bridge between digital and physical channels.

Impact: Eliminate stockouts, cut overstock 25%

Impact & Business Value

What happens if RetailX acts — and what's the cost of inaction?

Revenue at Risk	AOV Uplift Potential	Labor Efficiency Gain	Margin Recovery
~₹3.3L	**+10–15%**	**~20%**	**33.5%**
If top 5 customers churn	Via bundle promotions	By aligning shifts to Fri/Sun	Discount txns to be redesigned

Cost of Inaction

- Without a loyalty program, any single churned top customer removes ~₹65K–₹69K from annual revenue.
- Continuing flat discounts on 33.5% of transactions means permanently sacrificing margin with zero AOV improvement.
- Ignoring Fri/Sun peaks means understaffed stores during highest-traffic windows — directly hurting conversion.
- An unsynchronized omnichannel leads to stockouts online and overstock in-store — double the loss.

Limitations & Next Steps

⚠ Limitations

Small Customer Base:
n=25 unique customers severely limits churn prediction model generalizability.

Historical Data Only:
Analysis covers Jan 2022–Aug 2024; requires real-time streaming for operational pivots.

Simulated Dataset:
Synthetic data may not fully capture real-world demand volatility or outlier events.

No Demographic Data:
Customer ID-based analysis cannot reveal age, location, or psychographic segments.

Single-Store Context:
Findings may not generalize directly to multi-branch or franchise retail operations.

🚀 Next Steps

01 **Market Basket Analysis:**
Association rules to uncover cross-sell pairs (e.g. Butchers + Food).

02 **Predictive LTV Modeling:**
Forecast customer lifetime value for prioritized retention investment.

03 **Real-Time Streaming:**
Integrate Apache Kafka or AWS Kinesis for live sales monitoring.

04 **Churn Prediction Model:**
Build ML classifier once customer data is expanded beyond n=25.

05 **Extended Dataset:**
Include 2025 full-year data; add geo-location, demographic, and product SKU data.

Team & Contributions

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Strategy & Timeline, Tableau Dashboards

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Recommendations & Business Impact

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Report Integrity, Presentation QA

Thank you · RetailX Store Sales Analytics · April 29, 2026